综合练习

人员

许睿谦、阮文璋 到课

上周作业检查

上上周作业链接: https://vjudge.net/contest/816673


开始时间 : 2026-05-17 08:30 CST
☆ 2026-0517 六队上课 (综合练习)
结束时间 : 2026-12-11 16:30 CST
已开始 : 14:05:44:09
进行中
还剩余 : 194:02:15:50
概览 题目 提交状态 排名 (14:05:44:07) 讨论
偏好 克隆 更新 管理

Rank	Team	Score	Penalty	A 9 / 20	B 7 / 7	C 7 / 16	D 1 / 1	E 4 / 18
1	chenxinmiao (陈欣妙)	4	47179	7:00:48:00 (-2)	8:12:59:00	8:12:56:06 (-2)		8:13:36:00 (-2)
2	r111	4	50479	7:06:27:00 (-1)	9:12:51:00	9:12:48:00 (-2)		8:13:33:00 (-8)
3	yyycj	4	64748	7:00:51:13 (-1)	12:09:01:00	12:09:00:00	13:03:56:00	(-1)
4	lyc12345bc (刘奕辰)	4	68688	7:00:56:00 (-2)	13:13:31:00	13:13:30:00 (-3)		13:11:11:37
5	wcz123bc	3	42885	7:05:52:00		9:09:17:00		13:02:36:00 (-3)
6	lijinshu	3	49186	7:01:02:00 (-2)	13:12:43:00	13:12:41:00 (-2)		
7	junyan007008	3	58652	13:13:15:00 (-1)	13:13:59:00	13:13:58:34		
8	xuruiqian	2	40740	14:03:25:47	14:03:35:00			
9	ccs2012 (陈丛晟)	1	13736	9:12:56:36				

上周作业链接: https://vjudge.net/contest/818166

本周作业

https://vjudge.net/contest/820326 (课上讲了上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

课堂表现

同学们课上听讲都很认真, 这周的题目比起之前会相对简单一些, 同学们课后争取要把题目全部补完。

课堂内容

CF1902D Robot Queries

翻转之后会映射为一个特定的坐标, 只需要查原字符串在 l~r 中是否存在那个特定的坐标即可

```
#include <bits/stdc++.h>
#define x first
#define y second
```

```
using namespace std;

typedef pair<int,int> PII;
const int maxn = 2e5 + 5;
int xx[maxn], yy[maxn];
map<PII, vector<int>> mp;

void update(char c, int &x, int &y) {
  if (c == 'U') ++y;
  else if (c == 'D') --y;
  else if (c == 'L') --x;
  else ++x;
}

bool query(int l, int r, int x, int y) {
  auto it = lower_bound(mp[{x,y}].begin(), mp[{x,y}].end(), l);
  if (it!=mp[{x,y}].end() && *it<=r) return true;
  return false;
}

int main()
{
  int n, Q; cin >> n >> Q;
  string s; cin >> s; s = " " + s;
  int x = 0, y = 0; mp[{x,y}].push_back(0);

  for (int i = 1; i <= n; ++i) {
    update(s[i], x, y); mp[{x,y}].push_back(i);
    xx[i] = x, yy[i] = y;
  }

  while (Q -- ) {
    int x, y, l, r; cin >> x >> y >> l >> r;
    int dx = x - xx[l-1], dy = y - yy[l-1];
    if (query(0,l-1,x,y) || query(l,r,xx[r]-dx,yy[r]-dy) || query(r,n,x,y)) cout
<< "YES" << endl;
    else cout << "NO" << endl;
  }
  return 0;
}
```

CF463D Gargari and Permutations

可以转化为最长路问题

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1000 + 5, M = 5 + 5;
int p[M][N], deg[N], f[N];
```

```cpp
vector<int> vec[N];

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= m; ++i) {
    for (int j = 1; j <= n; ++j) {
      int x; cin >> x; p[i][x] = j;
    }
  }

  for (int i = 1; i <= n; ++i) {
    for (int j = 1; j <= n; ++j) {
      bool flag = true;
      for (int k = 1; k <= m; ++k) {
        if (p[k][i] >= p[k][j]) flag = false;
      }
      if (flag) vec[i].push_back(j), ++deg[j];
    }
  }

  queue<int> q;
  for (int i = 1; i <= n; ++i) {
    if (!deg[i]) f[i] = 1, q.push(i);
  }
  int res = 0;
  while (!q.empty()) {
    int u = q.front(); q.pop();
    res = max(res, f[u]);
    for (int i : vec[u]) {
      --deg[i];
      f[i] = max(f[i], f[u]+1);
      if (!deg[i]) q.push(i);
    }
  }
  cout << res << endl;
  return 0;
}
```

CF547B Mike and Feet

单调栈, 求每个数左右第一个比他小的即可

```cpp
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
int w[maxn], ll[maxn], rr[maxn], f[maxn];

int main()
```

```
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) cin >> w[i];

  stack<int> stk;
  for (int i = 1; i <= n; ++i) {
    while (!stk.empty() && w[i]<=w[stk.top()]) stk.pop();
    ll[i] = (stk.empty() ? 1 : stk.top()+1);
    stk.push(i);
  }

  while (!stk.empty()) stk.pop();
  for (int i = n; i >= 1; --i) {
    while (!stk.empty() && w[i]<=w[stk.top()]) stk.pop();
    rr[i] = (stk.empty() ? n : stk.top()-1);
    stk.push(i);
  }

  for (int i = 1; i <= n; ++i) {
    int len = rr[i] - ll[i] + 1;
    f[len] = max(f[len], w[i]);
  }
  for (int i = n; i >= 1; --i) f[i] = max(f[i+1], f[i]);

  for (int i = 1; i <= n; ++i) cout << f[i] << " ";
  cout << endl;
  return 0;
}
```

CF3B Lorry

枚举每个体积的物品选几个, 最后挑一种最好的情况

```
#include <bits/stdc++.h>
#define int long long

using namespace std;

const int maxn = 1e5 + 5;
struct node {
  int id, val;
  bool operator < (const node& p) const { return val < p.val; }
};
node w[3][maxn];
int p[3][maxn], len[3];

signed main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) {
    int a, b; cin >> a >> b;
```

```
    ++len[a]; w[a][len[a]] = {i,b};
  }

  for (int i = 1; i <= 2; ++i) {
    sort(w[i]+1, w[i]+len[i]+1); reverse(w[i]+1, w[i]+len[i]+1);
    for (int j = 1; j <= len[i]; ++j) p[i][j] = p[i][j-1] + w[i][j].val;
  }

  int res = 0, res1 = 0, res2 = 0;
  for (int i = 0; i <= len[2]; ++i) {
    if (i*2 > m) break;
    int j = min(len[1], m - i*2);
    int sum = p[2][i] + p[1][j];
    if (sum > res) res = sum, res1 = i, res2 = j;
  }

  cout << res << endl;
  for (int i = 1; i <= res1; ++i) cout << w[2][i].id << " ";
  for (int i = 1; i <= res2; ++i) cout << w[1][i].id << " ";
  cout << endl;
  return 0;
}
```

CF540D Bad Luck Island

f[i][j][k]: 最后每种分别剩 i/j/k 个时的概率是多少

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 100 + 5;
double f[maxn][maxn][maxn];

int main()
{
  int A, B, C; cin >> A >> B >> C;
  f[A][B][C] = 1;
  for (int i = A; i >= 0; --i) {
    for (int j = B; j >= 0; --j) {
      for (int k = C; k >= 0; --k) {
        double f1 = i*j, f2 = i*k, f3 = j*k, p = f[i][j][k];
        double fm = f1 + f2 + f3;
        if (i && k) f[i-1][j][k] += p * f2 / fm;
        if (i && j) f[i][j-1][k] += p * f1 / fm;
        if (j && k) f[i][j][k-1] += p * f3 / fm;
      }
    }
  }

  double res1 = 0, res2 = 0, res3 = 0;
```

```
  for (int i = 0; i <= A; ++i) {
    for (int j = 0; j <= B; ++j) {
      for (int k = 0; k <= C; ++k) {
        if (i && !j && !k) res1 += f[i][j][k];
        if (!i && j && !k) res2 += f[i][j][k];
        if (!i && !j && k) res3 += f[i][j][k];
      }
    }
  }

  printf("%.12f %.12f %.12f\n", res1, res2, res3);
  return 0;
}
```